EXPLORING THE CONSISTENCY OF MAXIMUM LIKELIHOOD ESTIMATOR OF IMPERFECT REPAIR $\mathbf{ARA_1}$ MODELS COMPUTED FROM A SINGLE TRAJECTORY*

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The simultaneous estimation of the three parameters of imperfect repair ARA_1 models by likelihood maximization remains unexplored yet, especially in the frequent industrial case when a single trajectory of failures is observed (following a single component). A theoretical approach to the consistency properties of the maximum likelihood estimator (MLE) is considered by exhibiting a multidimensional martingale based on the likelihood score, following a methodology initiated by Crowder (1975). Conditions of square integrability and in-law convergence of the martingale increments are studied, that highlight technical limitations required to ensure the consistency of the MLE. The multivariate mean value theorem is used to derive in-law convergence for the MLE from the martingale central limit theorem.

Keywords: imperfect repair model ; ARA_1 ; maximum likelihood ; martingale theory

1. Introduction

Let $\{T_i\}_{i\geq 1}$ be the successive failure times of a repairable system, starting from $T_0 = 0$. We assume that a repair task is performed after each failure

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and that repair durations are negligible. Let $\{X_i\}_{i\geq 1}$ be the times between failures ($X_i = T_i - T_{i-1}$) and N_t be the number of failures observed up to time t. Then, the failure process is a random point process. Its distribution can be characterized by the failure intensity, defined as:

$$\forall\, t \geq 0, \quad \lambda_t = \lim_{\Delta t \to 0} \frac{1}{\Delta t} P(N_{(t+\Delta t)-} - N_{t-} = 1 | \mathcal{H}_{t-}) \tag{1}$$

where $\mathcal{H}_{t-}$ is the history of the failure process just before time t, i.e. the set of all events occurred just before t. N_{t-} denotes the left-hand limit of N_t. In most cases, the failure process is a self-excited process, i.e. λ_t is a function of the number of failures and the failure times before t: $\lambda_t = \lambda_t(N_{t-}, T_1, \ldots, T_{N_{t-}})$. In this case, the distribution of the failure process is completely given by the intensity. It is assumed that before the first failure, the intensity is a deterministic continuous function of time, denoted $\lambda(t)$, and called the initial intensity. The initial intensity characterizes the intrinsic behaviour of a new unrepaired system. λ is the hazard rate of the first failure time $T_1 = X_1$. In the following, it is assumed that the initial intensity is the intensity of a Power Law Process (PLP):

$$\lambda(t) = \alpha\beta t^{\beta-1}, \ \alpha > 0, \beta > 0 \tag{2}$$

In general, the system is ageing intrinsically, so $\lambda(t)$ is increasing and $\beta > 1$.

In this article the Arithmetic Reduction of Age (ARA) class of imperfect repair models (based on the notion of virtual age proposed by Kijima 1989), defined in Doyen and Gaudoin (2004), is considered. The Arithmetic Reduction of Age model with memory m, denoted ARA_m, is defined by its failure intensity:

$$\lambda_t = \lambda \left(t - \rho \sum_{j=0}^{\min(m-1, N_{t-}-1)} (1-\rho)^j T_{N_{t-}-j} \right) \tag{3}$$

m reflects a Markovian property: it is the maximal number of previous failure times involved in the failure intensity. ρ is a parameter which characterizes repair efficiency. For $m = 1$, the failure intensity of the ARA_1 model is:

$$\lambda_t = \lambda \left(t - \rho T_{N_{t-}} \right). \tag{4}$$

In this model, the effect of a repair is to reduce the virtual age just before repair of a quantity proportional to the time elapsed since last repair: the

effective age is defined by

$$A_i = A_{i-1} + X_i - \rho X_i \Longrightarrow A_i = (1-\rho)T_i$$

The ARA_1 model is similar to Kijima's type I model (Kijima 1989) with deterministic repair effects.

2. Towards the convergence of the maximum likelihood estimator

2.1. *The likelihood*

Denote $\theta = (\alpha, \beta, \rho)$ the vector of unknown parameters. We do not consider a device still running, so that the current (right-censoring) time is not accounted for. We always assume that the system is submitted to ageing, so that $\beta > 1$. Denoting $t_0 = 0$, the likelihood of observed failures times $(t_1, \ldots, t_n)$ following an ARA_1 model is:

$$L_n(\theta) = \alpha^n \beta^n V_n^{\beta-1}(\rho) \exp\left\{-\alpha Z_n(\beta, \rho)\right\}$$

where

$$\begin{aligned} V_n(\rho) &= \prod_{i=1}^{n} \left(t_i - S_i(\rho)\right), \\ Z_n(\beta, \rho) &= \sum_{i=1}^{n} \left\{ \left(t_i - S_i(\rho)\right)^{\beta} - \left(t_{i-1} - S_i(\rho)\right)^{\beta} \right\}, \\ S_1(\rho) &= 0 \ \text{ and } \ \forall i \geq 2,\ S_i(\rho) = \rho t_{i-1}. \end{aligned}$$

Consequently:

$$Z_n(\beta, \rho) = \sum_{i=1}^{n} \left\{ \left(t_i - S_i(\rho)\right)^{\beta} - (1-\rho)^{\beta} t_{i-1}^{\beta} \right\}.$$

2.2. *Infering on θ*

To simplify the formalism, denote the gradient of the log-likelihood $\ell_n(\theta)$. The classical inference is to estimate θ by the maximum likelihood estimator (MLE) $\hat{\theta}_n$, defined as the solution of $\boldsymbol{\ell}'_n(\hat{\theta}_n) = 0$ where

$$\boldsymbol{\ell}'_{\mathbf{n}}(\theta) = \left(\frac{\partial \ell_n(\theta)}{\partial \alpha}, \frac{\partial \ell_n(\theta)}{\partial \beta}, \frac{\partial \ell_n(\theta)}{\partial \rho} \right)^T.$$

2.3. *Studying the behavior of $\hat{\theta}_n$*

Inspired by Crowder (1975), we consider the following property. Denote $\mathcal{F}_n$ the $\sigma-$algebra generated by n time steps of the considered process. Next proposition provides a classical result for regular models (Merkouris 2007), while algebraic calculus remains needed to provide expressions of martingale increments. It reveals that the usual regularity conditions (in the sense of Wald) needed to prove the convergence of maximum likelihood estimators are limited to cases where ageing is not only sure ($\beta > 1$) but strong too ($\beta > 2$), namely the speed of ageing is increasing through time.

Proposition 2.1. *Assume $\beta > 1$. Then the three-dimensional vector defined by $M_0 = 0$ and $\forall n \geq 1$,*

$$\mathbf{M_n}(\theta) = \boldsymbol{\ell}'_{\mathbf{n}}(\theta)$$

is a $\mathcal{F}_{n-1}-$adapted martingale. It is square integrable if and only if $\beta > 2$.

It must be noticed, however, that the loss of square integrability when $1 < \beta \leq 2$ is only due to the unknowledge of ρ. If ρ is set to a known value, regularity conditions are not limitative over (α, β), as previously found by some authors (Doyen et al. 2004). An application of the multivariate martingale central limit theorem states the following result.

Proposition 2.2. *If $\beta > 2$, there exists a positive invertible matrix $\Sigma(\theta)$ such that*

$$\frac{1}{\sqrt{n}} M_n(\theta) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \Sigma(\theta)).$$

Proving this proposition requires to solve several issues.

(1) Deriving the components of the covariance matrix $\mathbf{s_n} = \mathrm{E}\left[< \mathbf{M} >_\mathbf{n}\right]$ where $< \mathbf{M} >_\mathbf{n}$ is the multivariate increasing process of $(\mathbf{M_n})_n$ defined by the conditional expectancies of martingale increments:

$$\begin{aligned} < \mathbf{M} >_\mathbf{n} &= \sum_{k=1}^{n} \mathrm{E}\left[\boldsymbol{\Delta}\mathbf{M_k}(\theta)\boldsymbol{\Delta}\mathbf{M_k}(\theta)^T | \mathcal{F}_{k-1}\right], \\ &= \sum_{k=1}^{n} \begin{pmatrix} \mathrm{E}\left[\Delta_{1,k}^2 | \mathcal{F}_{k-1}\right] & \mathrm{E}\left[\Delta_{1,k}\Delta_{2,k} | \mathcal{F}_{k-1}\right] & \mathrm{E}\left[\Delta_{1,k}\Delta_{3,k} | \mathcal{F}_{k-1}\right] \\ \mathrm{E}\left[\Delta_{1,k}\Delta_{2,k} | \mathcal{F}_{k-1}\right] & \mathrm{E}\left[\Delta_{2,k}^2 | \mathcal{F}_{k-1}\right] & \mathrm{E}\left[\Delta_{2,k}\Delta_{3,k} | \mathcal{F}_{k-1}\right] \\ \mathrm{E}\left[\Delta_{1,k}\Delta_{3,k} | \mathcal{F}_{k-1}\right] & \mathrm{E}\left[\Delta_{2,k}\Delta_{3,k} | \mathcal{F}_{k-1}\right] & \mathrm{E}\left[\Delta_{3,k}^2 | \mathcal{F}_{k-1}\right] \end{pmatrix}. \end{aligned}$$

(2) Getting a multivariate Law of Large Number under the form

$$A_n M_n \xrightarrow[I\!P]{} I_3 \tag{5}$$

where $A_n \simeq s_n^{-1/2}$.

(3) Proving the Lindeberg condition

$$\frac{a_{i,j}}{n} \sum_{k=1}^{n} \mathrm{E}\left[\Delta_{i,k}\Delta_{j,k}\mathbb{1}_{\Gamma_{i,j,k}>\epsilon}|\mathcal{A}_{k-1}\right] \xrightarrow[I\!P]{} 0. \tag{6}$$

where $\Gamma_{i,j,k} = \frac{\alpha}{n}|\sqrt{\Delta_{i,k}\Delta_{j,k}}|$ and $a_{i,j}$ the (i,j) element of $a = \tilde{s}^{-1} = nA_n^2$.

From the multivariate mean value theorem, there exists $\bar{\theta}_n \in \Theta$ (provided Θ be an open convex set), defined by $\bar{\theta}_n = (1-\nu)\theta + \nu\hat{\theta}_n$ where $\nu \in [0,1]$, such that

$$\boldsymbol{\ell}'_n(\hat{\theta}_n) - \boldsymbol{\ell}'_n(\theta) = \boldsymbol{\ell}''_n(\bar{\theta}_n) \cdot \left(\hat{\theta}_n - \theta\right)$$

where $\boldsymbol{\ell}''_n(\theta)$ is the Hessian matrix and $\cdot$ is for the dot product. Because $\ell'_n(\hat{\theta}_n) = 0$ by definition, using Slutsky's theorem and Equations (5-6), a central limit theorem can be got:

$$\frac{\Sigma^{-1/2}(\theta) \cdot \boldsymbol{\ell}''_n(\bar{\theta}_n)}{\sqrt{n}} \cdot \left(\hat{\theta}_n - \theta\right) \sim \mathcal{N}(0, I_3).$$

The details of calculations and components of matrix $\Sigma(\theta)$ are too long to be detailed here. They will be explained during the conference talk.

3. Concluding remark

These first results will be detailed and pursued in a future article. But it already appears that most often, when $\beta < 2$, the current conditions under which the maximum likelihood estimation is carried out cannot guarantee a good and controllable behavior of the estimator. In such cases a regularizing technique, for instance based on Bayesian computation, should be preferred. It however requires objective reasoning for eliciting prior distributions. In other cases, especially those corresponding to situations where ageing is strong, our results prove the usefulness and validity of the use of classical theory, as well as the relevance of bootstrap techniques for estimating the uncertainty of the maximum likelihood estimator. Therfore our results give a theoretical basis to the current practice of applied statisticians in such situations.